

House Price Prediction – Regression Model Report

1. Methodology

The project used a supervised machine learning regression approach to predict house prices. First, the dataset was explored using descriptive statistics, distributions, skewness, correlation analysis, and an Actual vs Predicted plot. Highly skewed variables were identified for further consideration during preprocessing. The target variable was separated from the input features, and the data was divided into 80% training and 20% testing sets using `random_state=42`. For Linear Regression and Ridge Regression, numerical features were standardized using `StandardScaler`. Decision Tree Regression was trained on the original, unscaled features because tree-based models do not require feature scaling. Three models were trained and evaluated using MAE, MSE, RMSE, and R^2 .

2. Model Results

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.5332	0.5559	0.7456	0.5758
Ridge Regression	0.5332	0.5558	0.7455	0.5759
Decision Tree (depth=5)	0.5223	0.5245	0.7242	0.5997

3. Results Interpretation

Lower MAE, MSE, and RMSE indicate smaller prediction errors, while a higher R^2 indicates that the model explains more variation in house prices. The Decision Tree achieved the lowest MAE (0.5223) and RMSE (0.7242), and the highest R^2 (0.5997) among the three tested models. Linear and Ridge Regression produced almost identical results, showing that Ridge regularization with $\alpha=1.0$ made very little difference for this dataset. The Actual vs Predicted plot shows a clear positive relationship, but substantial scatter around the ideal diagonal line remains.

4. Conclusion

Among the three baseline models, the Decision Tree with `max_depth=5` performed best on the current test set. However, an R^2 of approximately 0.60 means that considerable variation in house prices remains unexplained. The prediction plot also indicates errors at both low and high house values, with the tree producing repeated prediction levels. The next improvement step should be hyperparameter tuning of the Decision Tree (for example, `max_depth`, `min_samples_split`, and `min_samples_leaf`), followed by comparison using the same test set or cross-validation. Feature engineering and appropriate treatment of strongly skewed variables may also improve performance.

Key takeaway: Decision Tree currently gives the strongest baseline performance, but further tuning and feature engineering are needed for a more accurate house-price model.